

Automated Leukaemia Cancer Detection Using Deep Learning Techniques

Gokul S
Department of ECE,
Bannari Amman Institute of
Technology
Tamilnadu, India
gokul.sengottaian1@gmail.com

Gokul shrinivas K
Department of ECE,
Bannari Amman Institute of
Technology
Tamilnadu, India
gokulshrinivas3@gmail.com

Vishnuvarshan R
Department of ECE,
Bannari Amman Institute of
Technology
Tamilnadu, India
vishnuvarshan23@gmail.com

Leeban Moses M
Department of ECE,
Bannari Amman Institute of
Technology,
Tamilnadu, India
leebanmoses@gmail.com

Abstract— Leukaemia, a critical blood cancer, is marked by an excessive proliferation of immature white blood cells in the bone marrow, significantly impacting healthy cell function and presenting serious health risks. Early and accurate diagnosis of leukaemia remains a major challenge, with the disease categorized into acute and chronic forms, and acute lymphocytic leukaemia (ALL) accounting for around 25% of childhood cancer cases. This paper proposes a novel method for leukaemia detection using advanced deep learning techniques. We specifically use VGG19 and ResNet152, to analyze blood smear images and extract detailed features that enhance classification precision. Our approach is novel in integrating deep learning models with traditional machine learning classifiers, including Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision Tree, Random Forest, XGBoost, Logistic Regression, and AdaBoost. This hybrid method leverages the strengths of both deep learning for feature extraction and machine learning classifiers for enhanced classification, setting it apart from previous approaches that rely solely on deep learning models. The experimental results highlight the effectiveness of this deep learning-based approach in improving diagnostic accuracy and supporting effective clinical decision-making.

Keywords— Leukaemia Detection, Deep Learning, VGG19, ResNet152, Hybrid Classification

I. INTRODUCTION

Cancer remains a leading cause of non-accidental deaths globally, with the World Health Organization (WHO) reporting around 10 million cancer-related deaths in 2020. Leukaemia, a particularly aggressive blood cancer characterized by abnormal white blood cell growth, accounted for 311,594 deaths out of 474,519 diagnosed cases that year. This high mortality rate underscores the urgent need for improved detection and treatment. Factors influencing leukaemia incidence include gender, age, genetic predisposition, and exposure to toxins. Leukaemia is categorized into subtypes like acute (ALL) mainly affects lymphoid cells and is prevalent in children, while (AML) targets myeloid cells [1]. Early detection and treatment are critical, as traditional microscopic examination of stained blood slides is time-consuming and dependent on pathologist expertise. This has spurred the development of automated diagnostic systems to address these limitations. [2,3].

Leukaemia is a form of blood cancer characterized by the excessive production of white blood cells. This abnormal increase disrupts the balance of blood components, as these cells overcrowd the essential red blood cells and platelets. Leukaemia is categorized into acute or chronic types and further classified into true or not, based on the originating cell lineage. The most common forms are acute myeloid leukaemia (AML) and chronic myeloid leukaemia (CML), both deriving from the myeloid lineage. These conditions involve the overproduction of blood cells by the bone marrow, leading to a range of complications like infections, fatigue, anemia, and bleeding disorders. As the number of abnormal cells grows, symptoms tend to worsen over time [4,5]. This study focuses on (ALL), utilizing it as the subject for developing advanced diagnostic methodologies. Microscopic image analysis remains a cornerstone in the early detection and accurate diagnosis of leukaemia. This analysis involves a detailed examination of blood samples under a microscope, assessing various parameters such as cell size, morphology, and the relative abundance of different blood cell types, including erythrocytes, leukocytes, and platelets. correct leukaemia diagnosis is critical for effective treatment. Conventional diagnostic techniques rely predominantly on manual microscopic inspection, which, while widely used, is inherently time-consuming and heavily dependent on the expertise of clinicians[7]. The subjective nature of this approach, coupled with the potential for prolonged examination periods, can lead to diagnostic delays, adversely impacting patient outcomes Automated diagnostic systems for leukaemia detection offer a promising alternative to traditional methods, reducing human intervention and providing more consistent, accurate results. Diagnosing Acute Lymphoblastic Leukaemia (ALL) is challenging due to its complexity and variability. Traditional methods, including physical exams and large-scale lab analyses, are intensive and costly, and may lead to delayed or inaccurate diagnoses. This can allow the disease to advance unnoticed. Machine learning (ML) has enhanced digital image analysis by uncovering complex patterns in medical images that human observation might miss. ML models outperform traditional methods in accuracy, cost-effectiveness, and processing speed. This research introduces a hybrid approach combining convolutional neural networks (CNNs) like VGG19 and ResNet152 with

various machine learning algorithms to develop an automated leukaemia detection system. The system extracts features from blood smear images using CNNs and classifies them with algorithms such as SVM, KNN, Decision Tree, Random Forest, XGBoost, Logistic Regression, and AdaBoost. The resulting model, integrating the most effective classifier with deep neural networks, demonstrates superior precision and speed in identifying leukaemia compared to traditional methods..

II. LITERATURE SURVEY

[6] proposed a computational deep learning approach for pattern recognition of Acute Lymphoblastic Leukaemia (ALL), a cancer affecting the blood and bone marrow. Their model, named ALLDM, uses pattern recognition algorithms to analyze large datasets and detect patterns associated with ALL. The study demonstrated high performance in various chemotherapy management and treatment categories. [7] reviewed CNN-based methods for medical image classification, analyzing 149 key papers. They examined techniques like data preprocessing, transfer learning, and CNN architectures, highlighting their role in improving accuracy. While CNNs have shown promise, the study also discussed challenges in clinical applications and suggested future research to enhance their integration into clinical practice. [8] used a Naive Bayes classification algorithm on the NCBI GEO dataset to predict Leukaemia in patients using gene features. Their methodology involved normalizing the dataset, calculating similarity scores for a fuzzy table, and applying a graph-based statistical model, resulting in high prediction accuracy. [9] explore various Machine Learning and Deep Learning techniques, particularly feature selection and classification. They compare algorithm performance, evaluating accuracy across different datasets, and incorporate additional methods to address specific challenges and identify key factors influencing the classification process. [10] demonstrated the effectiveness of the EfficientNet B3 architecture in classifying leukaemia subtypes, achieving 98.95% accuracy with a minimal loss of 0.120. Using the Kaggle Leukaemia cancer dataset, their research highlights the model's potential in early diagnosis and personalized treatment strategies for leukaemia. [11] present a study on classifying leukaemia using SVM classifiers. Leukaemia is categorized into acute and chronic types, with further subdivisions based on the affected blood cells. The analysis uses image processing techniques on blood smear images of healthy and leukemic individuals to differentiate between types of leukaemia. [12] A systems biology approach was used to elucidate gene regulatory networks for leukaemia markers, simplifying complex genetic interactions and highlighting crucial markers linked to leukaemia-associated genes [12]. [13] introduced a compressive sensing (CS) method for leukaemia subtyping, achieving 97% classification accuracy. ALL and AML using selected genes from 7129 samples. Their CS-based detector, validated on 38 subjects, demonstrates the potential of CS for improved leukaemia diagnosis. [14] proposed a leukaemia detection system utilizing Apache Spark's BigDL library and GoogleNet CNN architecture, achieving superior accuracy in detecting four types of leukaemia. Their model achieved 97.33% training accuracy and 94.78% validation accuracy, outperforming a traditional Keras model with notable improvements in efficiency and accuracy. [15] propose a machine learning-based approach for leukaemia detection, utilizing image

processing and transfer learning. By extracting morphological characteristics of lymphocytes, leukaemia is classified into AML, ALL, CML, or CLL using a hybrid ensemble model that combines Inception and Xception convolutional networks.

III. DATASET COLLECTION AND PROPOSED METHODOLOGY

In this study, we utilized a dataset of histopathological images of leukemic cancer, sourced from an accessible public repository. The dataset features a mix of benign and malignant leukemic tissue images, providing a balanced set of examples for classification purposes. The leukaemia dataset shown focuses on human subjects, specifically from blood and bone samples. It includes data from 118 subjects and consists of histopathology data types. The dataset is related to leukaemia, a type of cancer, with a total size of 10.44 GB. Its status is marked as public and complete, making it accessible for use. The dataset was last updated on March 26, 2019, providing valuable insights for leukaemia research and analysis from C-NMC 2019 | C_NMC_2019 Dataset: ALL Challenge dataset of ISBI 2019. Before Feature extraction we were involved in the preprocessing phase such as resizing the images to a uniform dimension of 512×512 pixels. This standardization is essential for ensuring compatibility with the deep learning models used in our analysis. Once resized, the images were split into two subsets: 80% for training the models and 20% for testing their performance. This division allows us to train the models effectively while reserving a portion of the data for evaluating their accuracy and generalizability. After that, deep learning models were chosen with suitable machine learning classifiers. For the deep learning aspect of our study, we employed several well-known models. We started with VGG19, a deep convolutional neural network renowned for its ability to capture detailed image features.

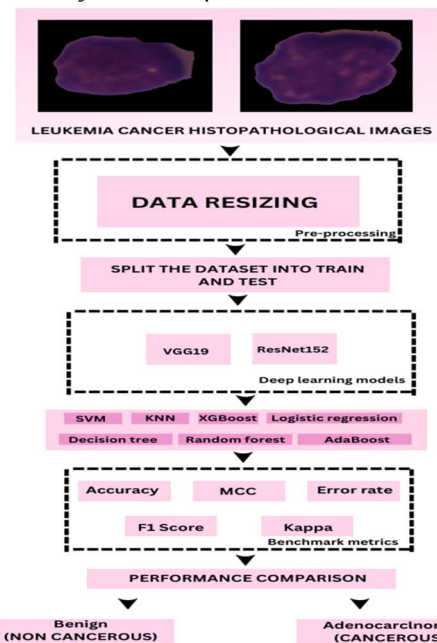


Fig. 1. Methodology proposed for Leukaemia detection

Additionally, we utilized ResNet-152, which incorporates a residual learning framework to tackle challenges such as vanishing gradients and improve feature extraction in very deep networks. To compare results, we also included a less complex version of ResNet-152, to

assess how shallower networks perform relative to deeper ones. To enhance our classification results, we also applied several traditional machine learning classifiers. These classifiers were trained on features extracted by deep learning models to evaluate their effectiveness in differentiating between benign and malignant leukemic tissues. We assessed model performance using metrics to thoroughly compare and identify the most effective approach. Our goal was to accurately classify images as either benign or adenocarcinoma, combining deep learning and traditional methods for optimal detection. The overall methodology, from image preprocessing to classification, in “Fig. 1”.

IV. FEATURE EXTRACTION

Feature extraction is key to improving accuracy in leukaemia cancer detection. This study compares traditional methods with deep learning techniques using ResNet-152 and VGG19 models. These CNNs extract important features from blood smear images for effective leukaemia classification.

A. VGG19

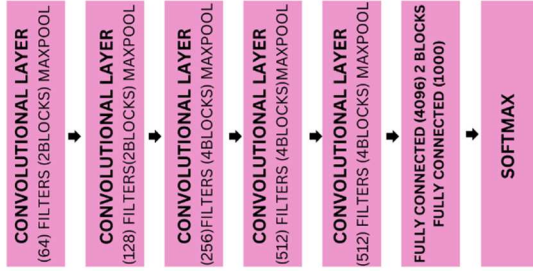


Fig. 2. VGG19 Architecture

The “Fig. 2”, VGG19 is a deep learning model renowned for image classification due to its deep yet simple architecture. It features 19 layers—16 convolutional and 3 fully connected—allowing for detailed feature extraction. We applied VGG19 to analyze blood smear images to accurately identify leukaemia cells. The model uses small 3x3 filters in convolutional layers to capture fine details, with early layers focusing on basic features and deeper layers on complex patterns. Max pooling layers reduce feature map sizes, balancing efficiency with feature retention. Batch normalization and dropout layers were used to enhance training and prevent overfitting, making VGG19 effective in extracting detailed features from blood smear images.

B. ResNet152

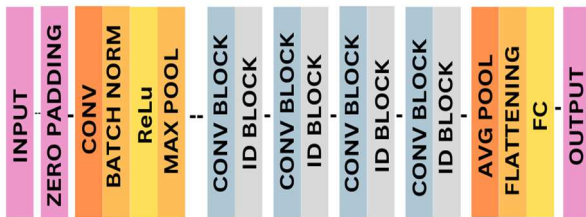


Fig. 3. ResNet152 Architecture

The “Fig. 3”, ResNet152 is a powerful deep learning model known for its innovative residual learning approach, allowing it to train very deep networks without vanishing gradient issues. Comprising 152 layers, it starts with a 7x7

convolutional layer and a max pooling layer to reduce spatial dimensions. The core consists of residual blocks with shortcut connections that bypass layers, enabling the model to learn residual functions efficiently. ResNet152 integrates batch normalization and ReLU activation functions to stabilize and streamline training. Instead of fully connected layers, it uses a global average pooling layer at the end to reduce overfitting by averaging features before classification. This architecture’s depth and efficiency make ResNet152 ideal for tasks like leukaemia detection, where detailed feature capture is essential.

V. CLASSIFICATION WITH ML ALGORITHMS

This study uses classifiers like SVM, KNN, XGBoost, Random Forest, and Decision Tree to detect leukaemia. Classification is the final step in medical image analysis, where patterns are identified and labeled. It involves training the model with data and then testing it to see if images are benign or malignant, excluding those that don't fit any category.

A. Adaboost

(Adaptive Boosting) is an ensemble learning technique designed to improve both classification and regression tasks by sequentially training weak classifiers. Each subsequent classifier places more emphasis on the data points that were incorrectly classified in earlier iterations. The final model is a weighted combination of all classifiers, with greater weight assigned to those with higher accuracy given I Eq. (1).

$$y_j = \sum_{n=1}^n \alpha_n k_n(x_j) \quad (1)$$

B. Decision Tree

Decision Trees (DT) utilize a hierarchical structure to recursively split data into decision points, represented by internal nodes, and final classifications, represented by leaf nodes. The process begins by selecting the most important variable for the root node, followed by dividing the data according to criteria such as entropy, Gini index, or information gain given in Eq. (2) and Eq. (3).

$$Gini(D_t) = 1 - \sum_{j=1}^m p_m^2 \quad (2)$$

Entropy:

$$E(D_t) = - \sum_{j=1}^m p_j \log_2(p_j) \quad (3)$$

C. KNN

K-Nearest Neighbors (KNN) is a simple yet effective supervised classification algorithm. It labels new test data by calculating the distance between test points and labeled training data, employing metrics such as Euclidean, Manhattan, or Minkowski distances. The classification of the test data is determined by the majority vote among its K closest neighbors, enabling efficient classification based on similarity shown in Eq. (4)

$$dist(a, b) = \sqrt{\sum_{j=1}^m (a_j - b_j)^2} \quad (4)$$

D. Logistic Regression

Logistic regression begins with input data of features and labels, extracting relevant features to identify patterns. These features are linearly combined using weights to produce a weighted sum, which is then passed through a sigmoid function. The sigmoid converts the sum into a probability between 0 and 1, indicating the likelihood of the instance belonging to the positive class. A threshold is applied to classify the instance as positive or negative, and the final output is the predicted class for each input.

E. SVM

Its is a supervised learning algorithm that uses kernel functions, such as polynomial, RBF, Gaussian, and linear, to map data into a higher-dimensional space, enabling linear separation. By finding the optimal hyperplane in this space, SVM effectively handles non-linear decision boundaries, allowing for accurate classification of inputs given in Eq. (5)

$$w \cdot a + b = 0 \quad (5)$$

F. XGBoost

XGBoost is a robust and scalable gradient boosting library designed for high efficiency. It boosts predictive performance by combining several weak models and is adept at handling large datasets. Notable features include its capability to address missing values and its support for parallel processing, making it highly effective for real-world scenarios given in Eq. (6).

$$\hat{x}_k = \sum_{l=1}^L f_m(x_k) \quad (6)$$

G. Random forest

classifier merges the results from several decision trees, where each tree casts a vote for the most frequent class. This method boosts classification accuracy by efficiently dealing with high-dimensional data and multicollinearity, all while preserving computational speed and reducing the risk of overfitting given in Eq. (7) and Eq. (8).

$$\hat{a}_y = \text{mode}(g_1(b_y), g_2(b_y), g_3(b_y), \dots, g_x(b_y)) \quad (7)$$

For regression

$$\hat{a}_y = \frac{1}{z} \sum_{z=1}^z g_z(x_y) \quad (8)$$

VI. RESULTS AND DISCUSSION

we outline the performance metrics, experimental results, and comparative analysis of the deep learning models employed for leukaemia detection. We evaluated each model using critical metrics, including accuracy, error rate, Jaccard index, Matthews Correlation Coefficient (MCC), F1-score, G-mean, and Kappa statistic. These metrics, derived from True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), are vital for assessing model performance. Accuracy indicates the proportion of correct predictions, while error rate shows the proportion of incorrect predictions. The F1-score balances precision and

recall, the MCC evaluates binary classification by analyzing correlations between actual and predicted outcomes ,

The Kappa statistic measures the agreement between model predictions and actual outcomes, accounting for chance shown in Eq. (9).

$$kappa = \frac{p_o - p_e}{1 - p_e} \quad (9)$$

The G-Mean is particularly valuable for handling imbalanced datasets, as it reflects the model's performance across all classes, including those that are underrepresented given in Eq. (10).

$$G - Mean = \sqrt{\frac{TP + TN}{(TP + FN)(TN + FP)}} \quad (10)$$

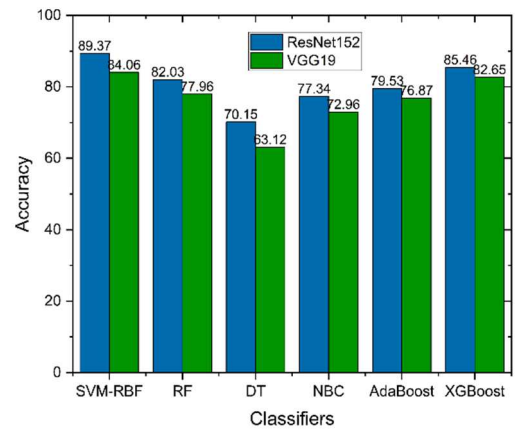


Fig. 4. Comparison of Accuracy

The “Fig. 4” clearly show that ResNet152 consistently outperforms VGG19 across all classifiers. For the SVM-RBF classifier, ResNet152 achieves the highest accuracy at 89.37%, while VGG19 scores 84.06%. With the Random Forest (RF) classifier, ResNet152 reaches 82.03% compared to VGG19's 77.96%. In Decision Trees (DT), ResNet152 posts 70.15%, significantly above VGG19's 63.12%. For the Naive Bayes Classifier (NBC), ResNet152 records 77.34%, with VGG19 at 72.96%. AdaBoost shows ResNet152 at 79.53%, ahead of VGG19's 76.87%, and in XGBoost, ResNet152 leads with 85.46%, while VGG19 scores 82.65%. Overall, ResNet152 demonstrates superior performance across all classifiers.

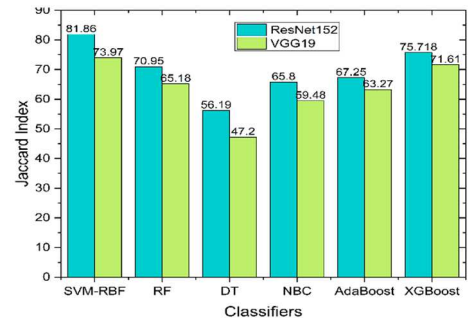


Fig. 5. Comparison of Jaccard Index

The “Fig. 5” shows the comparison of the Jaccard Index reveals a similar trend, where ResNet152 outperforms VGG19 in most cases. For the SVM-RBF classifier, ResNet152 achieves 81.86%, surpassing VGG19’s 73.97%. With the RF classifier, ResNet152 records a Jaccard Index of 70.95%, while VGG19 follows at 65.18%. A notable difference is observed in the DT classifier, where ResNet152 attains 56.19%, in contrast to VGG19’s lower 47.2%. For NBC, ResNet152 achieves a Jaccard Index of 65.8%, with VGG19 slightly trailing at 59.48%. AdaBoost presents a Jaccard score of 67.25% for ResNet152, while VGG19 achieves 63.27%. In the XGBoost classifier, ResNet152 scores 75.718%, while VGG19 attains 71.61%. These results suggest that ResNet152 offers better segmentation overlap, consistently higher across all classifiers.

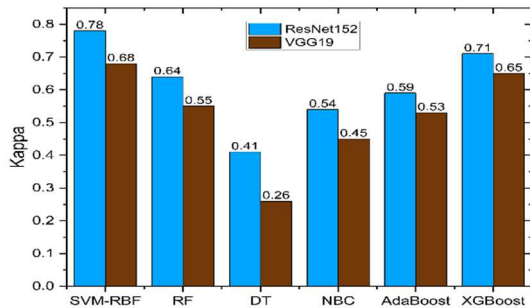


Fig. 6. Comparison of Kappa

In “Fig. 6” shows The Kappa scores further solidify ResNet152’s superiority over VGG19 in classifier performance. With the SVM-RBF classifier, ResNet152 achieves a Kappa score of 0.78, compared to VGG19’s 0.68. For the RF classifier, ResNet152 records 0.64, again outperforming VGG19, which scores 0.55. The DT classifier shows the largest discrepancy, with ResNet152 achieving 0.41, while VGG19 lags at 0.26. In the case of NBC, ResNet152 reaches 0.54, with VGG19 at 0.45. AdaBoost reveals a Kappa score of 0.59 for ResNet152. %. These results suggest that ResNet152 offers better across all classifiers.

In “Fig. 7” shows The F1-Score analysis reinforces the superiority of ResNet152 over VGG19. When utilizing the SVM-RBF classifier, ResNet152 achieves an impressive F1-Score of 90.02%, compared to VGG19’s 85.04%. In the RF classifier, ResNet152 scores 83.01%, while VGG19 follows with 78.92%. For DT, ResNet152 attains an F1-Score of 71.95%, significantly outperforming VGG19’s 64.13%. NBC also shows better results for ResNet152, which achieves an F1-Score of 79.37%, compared to VGG19’s 74.59%. AdaBoost exhibits an F1-Score of 80.41% for ResNet152, while VGG19 follows with 77.5%. Lastly, the XGBoost classifier yields an F1-Score of 86.18% for ResNet152, whereas VGG19 achieves 83.45%. These results indicate that ResNet152 consistently delivers more balanced and precise classifications, as evidenced by the higher F1-Scores across all classifiers.

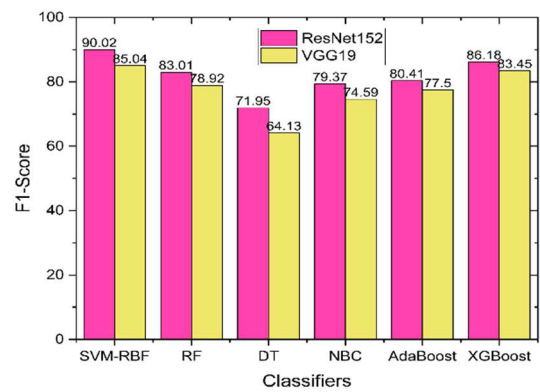


Fig. 7. Comparison of F1-Score

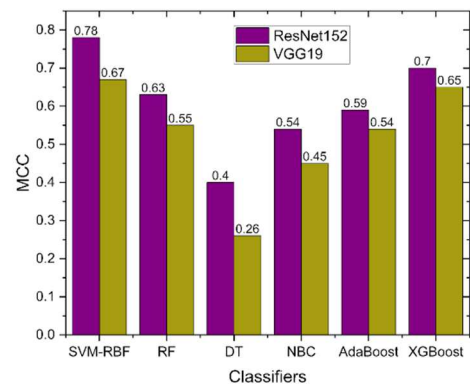


Fig. 8. Comparison of MCC

In “Fig. 8” offers a comparative analysis of the VGG19 and ResNet152 models using the Matthews Correlation Coefficient (MCC) across six classifiers. ResNet152 consistently outperforms VGG19, especially with the SVM-RBF classifier, achieving an MCC of 0.78 compared to VGG19’s 0.67, showing better handling of nonlinear patterns. In Random Forest, ResNet152 scores 0.63, surpassing VGG19’s 0.55, reflecting its stronger performance in ensemble techniques. Both models perform worse with Decision Trees, where ResNet152 records 0.40 and VGG19 0.26. ResNet152 also leads in Naive Bayes (0.54 vs. 0.45) and AdaBoost (0.59 vs. 0.54), and maintains its advantage in XGBoost with 0.70 over VGG19’s 0.65. %. These results suggest that ResNet152 offers better across all classifiers.

VII. CONCLUSION ANF FUTURE WORK

In conclusion, the comparison between VGG19 and ResNet152 in classifying acute lymphoblastic leukaemia (ALL) has provided valuable insights into their performance. ResNet152 proved to be the most effective model, achieving an accuracy of 89.38%, a Matthews Correlation Coefficient (MCC) of 0.787, a Kappa score of 0.787, and an F1-Score of 90.03%. These metrics demonstrate ResNet152’s ability to handle complex tasks with precision and reliability. While VGG19 also performed well with an accuracy of 84.06%, it was not as effective in capturing intricate patterns, particularly in comparison to ResNet152, which excelled in providing balanced and accurate predictions. This makes ResNet152 the preferred model for tackling complex medical image classification problems like leukaemia

detection. The novelty of this work lies in combining deep learning models, like ResNet152 and VGG19, with traditional machine learning classifiers (e.g., Random Forest, Logistic Regression, and SVM) for improved feature extraction and classification. This hybrid approach enhances both performance and computational efficiency, differentiating our study from traditional single-model approaches. For future work, we plan to expand the dataset to include more leukaemia subtypes, explore real-time clinical deployment, and incorporate explainable AI techniques for greater model transparency. These improvements will enhance the model's practicality and accuracy in real-world healthcare applications.

REFERENCES

- [1] J. Y. S. Lim, S. Bhatia, L. L. Robison, and J. J. Yang, "Genomics of racial and ethnic disparities in childhood acute lymphoblastic leukaemia," *Cancer*, vol. 120, no. 7, pp. 955-962, Apr. 2014 .
- [2] S. Dhalla, A. Mittal, and S. Gupta, "LeukoCapsNet: a resource-efficient modified CapsNet model to identify leukaemia from blood smear images," *Neural Comput. Appl.*, vol. 36, no. 5, pp. 2507-2524, Feb. 2024.
- [3] S. Gupta, N. Janu, and N. Shrotriya, "Leukaemia insight: Illuminating current diagnoses and forecasting futures with machine learning," in *Proc. 2024 1st Int. Conf. Computation of Artificial Intelligence and Machine Learning (ICCAIML)*, vol. 2184, Jaipur, Jan. 2024 .
- [4] S. Huang, Z. Chen, S. Zhong, Y. Zhang, C. Zeng, X. Zheng, Y. Li, and S. Chen, "Inhibition of TOX exerts anti-tumor effects in acute myeloid leukaemia by upregulating IRF7 expression," *Eur. J. Pharmacol.*, vol. 2024.
- [5] S. Ikeda, M. Suzuki, S. Tsunoda, and M. Ohta, "Dynamic changes in femoral bone marrow MRI in patients with chronic myelogenous leukaemia," *Int. J. Hematol.*, vol. 120, no. 3, pp. 269-270, Sep. 2024.
- [6] N. Jiwani, K. Gupta, G. Pau, and M. Alibakhshikenari, "Pattern recognition of acute lymphoblastic leukaemia (ALL) using computational deep learning," *IEEE Access*, vol. 11, pp. 29541-29553, 2023.
- [7] C. Chen, N. A. Mat Isa, and X. Liu, "A review of convolutional neural network based methods for medical image classification," *Comput. Biol. Med.*, vol. 185, article 109507, Feb. 2025.
- [8] Ul Alam, Efthakhar, Banik, Shovan, and Chowdhury, Linkon, "A statistical approach to classify the leukaemia patients from generic gene features," *Proc. 2020 ICCCI*, Coimbatore, India, Jan. 2020, pp. 9104106.
- [9] Divyapreethi B., Mohanarathinam A., "Analysis of Automated Leukaemia Cancer Detection using Feature Selection and Classification Techniques," *Proc. 2023 7th Int. Conf. I-SMAC (IoT in Social, Mobile, Analytics and Cloud)*, Kirtipur, Nepal, Oct. 2023, pp. 740-745 .
- [10] S. Afrin, S. K. Paul, R. R. Paul, M. M. Haque, M. S. Kabir, S. Fahim, and M. K. Isl, "Leukaemia Cancer Classification from Abnormal Cell Images Using a Deep Learning Approach: A Comprehensive Study," *Proc. 2023 26th Int. Conf. Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, Dec. 2023.
- [11] T. Dharani and S. Hariprasath, "Diagnosis of Leukaemia and its types Using Digital Image Processing Techniques," *Proc. 2018 3rd IEEE Int. Conf. Communication and Electronics Systems (ICCES)*, Coimbatore, India, Oct. 2018, pp. 275-279.
- [12] D. L. Tong and G. R. Ball, "Exploration of leukaemia gene regulatory networks using a systems biology approach," *Proc. 2014 IEEE Int. Conf. Bioinformatics and Biomedicine (BIBM)*, Belfast, United Kingdom, Nov. 2014, pp. 68-73.
- [13] W. Tang, H. Cao, and Y.-P. Wang, "Subtyping of leukaemia with gene expression analysis using compressive sensing method," *Proc. 2011 1st IEEE Int. Conf. Healthcare Informatics, Imaging and Systems Biology (HISB)*, New Orleans, LA, USA, Jul. 2011, pp. 76-80.
- [14] W. Tang, H. Cao, and Y.-P. Wang, "Subtyping of leukaemia with gene expression analysis using compressive sensing method," *Proc. 2011 1st IEEE Int. Conf. Healthcare Informatics, Imaging and Systems Biology (HISB)*, New Orleans, LA, USA, Jul. 2011, pp. 76-80 .
- [15] R. S. Rajeswari, C. S. Vasanth, C. Bhavana, and K. S. S. Chowdary, "Detection and Classification of Various Types of Leukaemia Using Image Processing, Transfer Learning and Ensemble Averaging Techniques," *Proc. 2022 2nd Asian Conf. Innovation in Technology (ASIANCON)*, Ravet, India, Aug. 2022 .